

Matching every DREAM trigger to an n_TOF wall + plastic coincidence

The calibration, what it delivers, and why it is not fitting itself

n_TOF X17 / DREAM analysis

2026-07-30

run_79 (stat090_0000 + stat090_0001) $\leftrightarrow$ n_TOF run 224572, v12_liqpileup reprocessing
2061 bunches, **213 420** non-flash DREAM triggers — the complete reference pair

The question, and the answer

Asked:

- ▶ For each DREAM trigger, is there an n_TOF SiPM-wall (top + bottom sum) coincident with a plastic pulse?
- ▶ Are the n_TOF detectors aligned in time?
- ▶ What are the miss and fake rates, and how do they depend on the search window?

Answered:

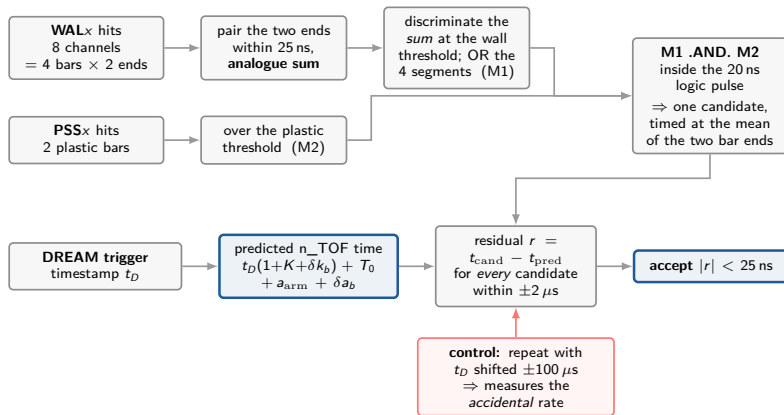
- ▶ **96.0 %** of triggers have one. The missing 4 % is 2.6 % plastic leg + 1.4 % wall leg.
- ▶ Yes — to a few ns everywhere, measured in situ on the file being analysed. The liquids need no offset at all.
- ▶ Once the DREAM clock is calibrated *per bunch*, the match residual is **flat at 6 ns** over the whole 80 ms, and **± 25 ns** is enough.

at $ r < 25$ ns	wall & plastic	wall only
efficiency	95.84 %	98.30 %
accidental	0.049 %	1.04 %
purity	99.998 %	99.982 %
≥ 2 candidates	2.46 %	5.36 %
≥ 2 arms	0.15 %	3.04 %

The accidental rate is *measured, not modelled*: the identical match with the DREAM time shifted $+100 \mu\text{s}$. The $-100 \mu\text{s}$ control agrees (0.062 %).

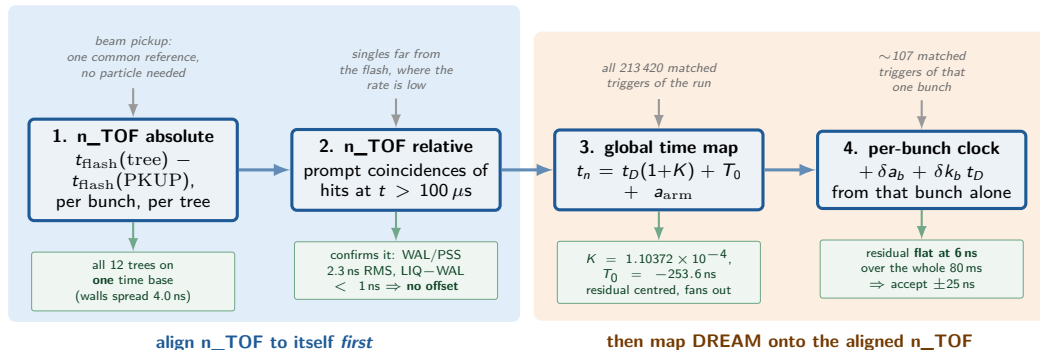
Everything on this deck is the **whole** reference pair, and every width quoted for the per-bunch fit is **cross-validated**.

What is being matched



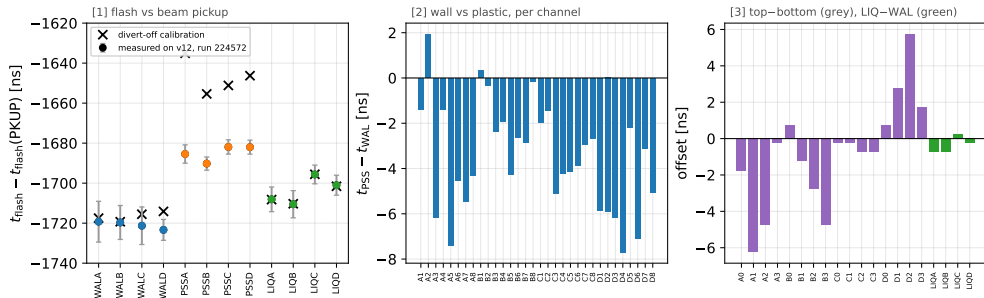
The DREAM trigger was the N1081B **sector SINGLES**, so that is what is rebuilt — not “any hit”. Thresholds are the values the discriminators actually held, read from each sub-run’s `n1081b_config.json`. The plastic requirement removes 92 % of the candidates (12 500 → 1 030 per bunch) for 2.6 % of the efficiency — and it is why the match is clean at 1–3 ms, where the wall singles rate alone would match everything by accident.

How the calibration is built: n_TOF to itself first, then DREAM onto it



The order matters. Steps 1–2 use only n_TOF information, so they cannot be contaminated by anything on the DREAM side; steps 3–4 then have a single, already-consistent n_TOF time base to map onto. Step 1 is the only one that can compare *different* detectors — it references everything to the beam pickup, which needs no common particle — and step 2 is the independent check that it worked.

Steps 1–2: the n_TOF detectors are aligned, measured in situ



[1] Absolute, vs the beam pickup.

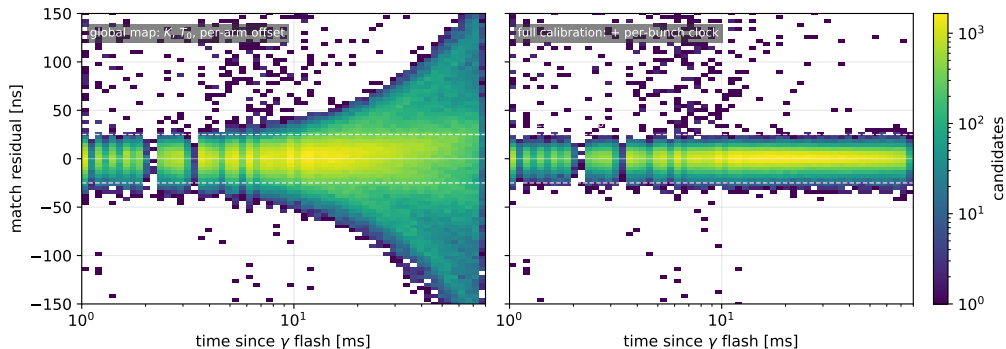
$t_{\text{flash}}(\text{tree}) - t_{\text{flash}}(\text{PKUP})$ per bunch. Walls $-1719.3 \dots -1723.3$ ns (spread **4.0 ns**). **The liquids reproduce the divert-off calibration to 0.1–0.5 ns** — two independent measurements agreeing at half a nanosecond. The plastics sit 31–50 ns from it, exactly as `flash_timing` warns: take those per run.

[2] Wall vs plastic, prompt coincidences of late hits ($t > 100 \mu\text{s}$): peak $-3.8 \dots -8.8$ ns per arm, $\sigma \simeq 11$ ns; per wall channel RMS **2.3 ns**, per plastic bar RMS 1.8 ns.

[3] Top vs bottom of a bar: within ± 6 ns on every segment. *Measure it on the file you analyse* — it is a property of the pulse reconstruction, not of the cabling.

[4] Liquid vs wall: $-0.8 \dots +0.2$ ns — **aligned to under a nanosecond**, so the liquid leg enters the merge with no offset.

Steps 3–4: the DREAM clock, globally and then per bunch

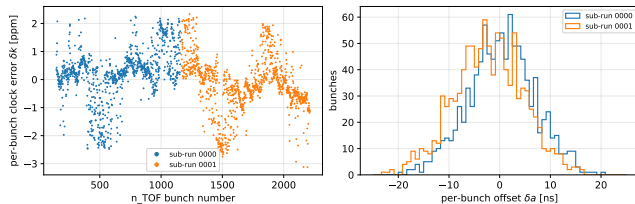


$r = t_{n_TOF} - t_{pred}$ for every candidate within $\pm 2 \mu s$ of the prediction.

Left, after the global map (K , T_0 , per-arm offset): centred, but the band **fans out in proportion to the time since the flash** — ± 9 ns at 1 ms, ± 37 ns at 40–80 ms. A width proportional to elapsed time is a *rate error*, not a timing resolution.

Right, after the per-bunch clock: flat, and it fits inside ± 25 ns at every time.

The DREAM timestamp clock wanders by about 1 ppm between bursts



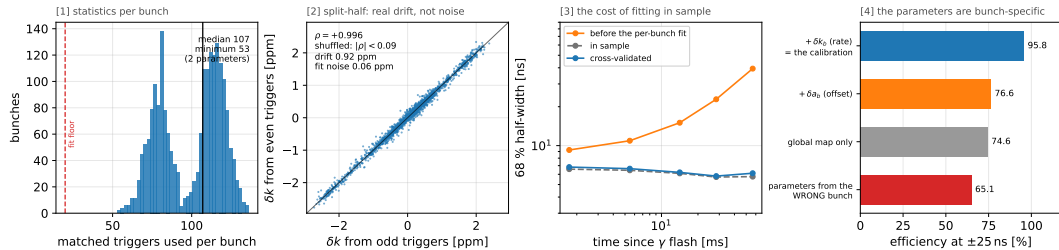
Fit, per bunch, from that bunch's own
~107 matched triggers:

$$t_{n_TOF} = t_{DREAM}(1+K+\delta k_b) + T_0 + a_{arm} + \delta a$$

- δk : RMS **0.92–0.96 ppm**, and *structured* in bunch number — neighbouring bursts drift together. 1 ppm is 60 ns at 60 ms, which is the whole fan on the previous slide.
- δa : RMS 6.5–6.8 ns.
- Fitted for **2061 of 2061** bunches.

Two free parameters against ≥ 53 matched triggers per bunch, so the correction itself is determined to **0.73 ns** — against a 6 ns residual.

Is the per-bunch fit reading back its own input? Four tests



The worry: the clock is fitted on triggers that were matched, and then used to match — a 2-parameter fit that chased noise would pull the prediction towards whatever candidate was nearest, narrowing the residual for free.

- [1] 107 triggers per bunch** (minimum 53) for 2 parameters. A fit absorbs $2/N \simeq 2\%$ of the variance, i.e. 1% of a width.
- [2] Fit each bunch *twice***, on its odd and its even triggers: $\rho = +0.996$, against $|\rho| < 0.09$ on 200 shuffles. Splitting the variance, **0.92 ppm of real drift against 0.06 ppm of fit noise**.
- [3] In-sample vs cross-validated width:** 6.5 vs 6.8 ns. *That gap is the overfitting*, measured — and everything quoted here is the cross-validated number.
- [4] Give each bunch *another* bunch's parameters** and the efficiency falls to 65%, *below* the uncorrected 74.6%. The fit is carrying bunch-specific truth, not a generic shape.

The decisive test. In a window far wider than the whole drift envelope the correction cannot change *who* is matched — only where inside the window they land. So if it were inventing partners, the efficiency at ± 500 ns would move:

efficiency at ± 500 ns	
global map only	96.0405 %
+ per-bunch clock	96.0405 %
difference	+0.0000 points

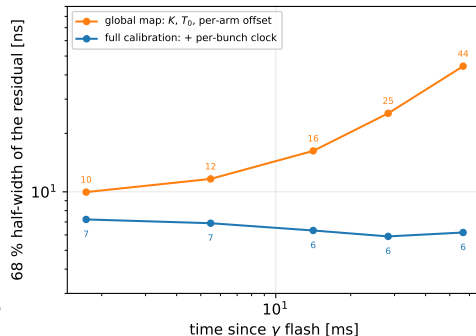
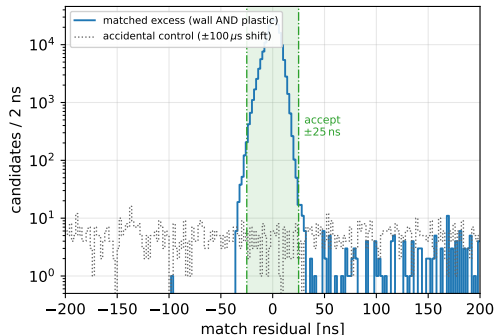
Identical to five decimal places. The per-bunch fit **concentrates** matches; it does not create them. What ± 25 ns then buys is background rejection.

And the sample it is fitted on is clean.

The complementary test — fit the clock on the *accidental* stream and see what it can invent — **cannot be run**, and that is the answer: a bunch has **0.4–0.5** accidental candidates within ± 200 ns of its predictions, against the 20 the fit requires. The population the clock is fitted on is 99.95 % real coincidences; there is no accidental component there to fit.

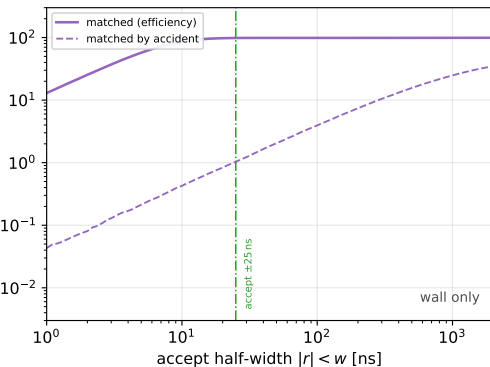
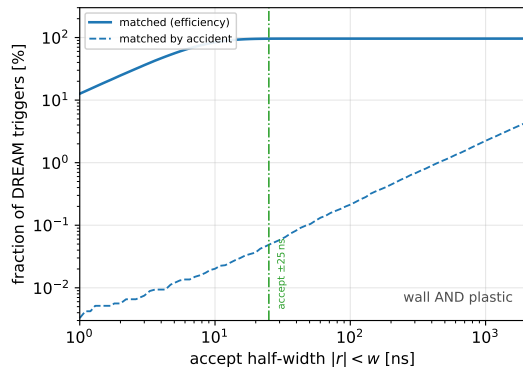
Consistently, applying the correction leaves the *measured* accidental rate where it was: 0.049 % $\rightarrow$ 0.049 % at ± 25 ns, unchanged to < 0.003 points on either sub-run.

The residual, and the window it allows



Left: the matched excess over the accidental control. A clean single band — no satellite, no tail structure.
Right: its 68 % half-width against time since the flash, before and after the per-bunch step of the calibration: $10 \rightarrow 7 \text{ ns}$ at 1–3 ms and $44 \rightarrow 6 \text{ ns}$ at 40–80 ms, i.e. **flat at $\simeq 6 \text{ ns}$ across the whole 80 ms**. That 6 ns is the real DREAM $\leftrightarrow$ n_TOF match resolution.

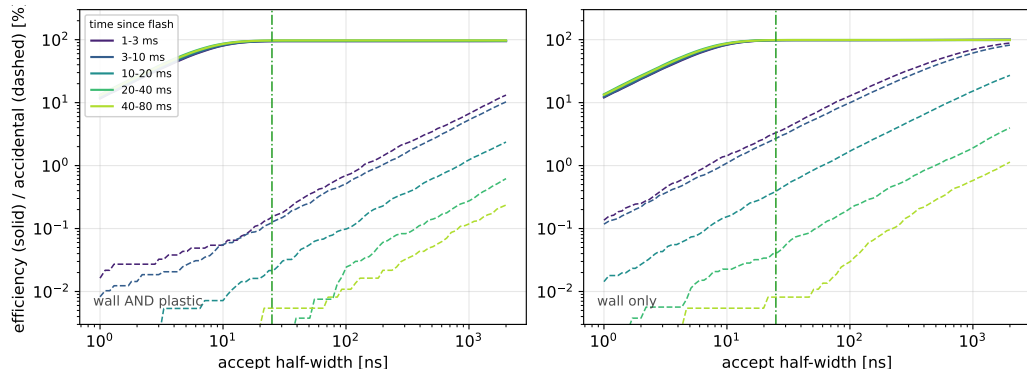
Misses and fakes versus the search window



Solid: the fraction of DREAM triggers matched. Dashed: the same with the DREAM time shifted $+100\ \mu\text{s}$, i.e. the rate of matching by accident.

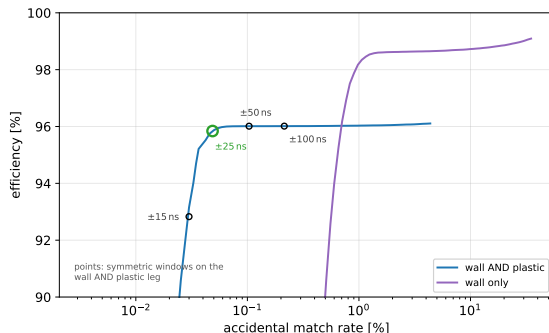
The accidental rate is *linear* in the window width — it is a Poisson singles rate — while the efficiency is flat from ~ 25 ns outward. Every nanosecond of window beyond the knee is therefore pure background, bought for nothing.

The same, per time since flash



The n_{TOF} singles rate falls by $\sim 100\times$ from 1–3 ms to 40–80 ms, so the accidental curves are stacked over two decades — but the *efficiency* curves lie on top of each other. **One window is right at all times:** the knee sits at 23.6 ns in every time bin and on both legs (28 ns in the single hardest bin, wall-only at 1–3 ms). This is what the per-bunch calibration buys: without it the knee would move with time since flash, and no single window could serve the whole spill.

The operating point: ± 25 ns, one band

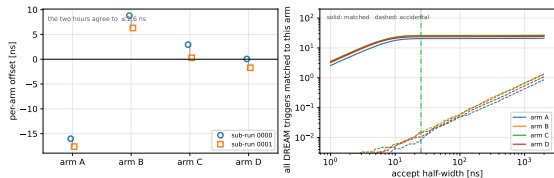


Criterion: the *tightest* window still within 0.5 % (relative) of the efficiency plateau. It lands at 23.6 ns $\Rightarrow$ quote ± 25 ns.

wall & plastic	eff	accid.	purity
1–3 ms	94.4 %	0.147 %	99.991 %
3–10 ms	94.7 %	0.128 %	99.993 %
10–20 ms	95.7 %	0.022 %	99.999 %
20–40 ms	96.8 %	0.002 %	99.9999 %
40–80 ms	96.9 %	0.005 %	99.9998 %
all	95.8 %	0.049 %	99.998 %

Going wider is not free and does not pay: ± 150 ns costs 7 $\times$ the background for +0.17 points of efficiency, and nearly doubles the two-arm ambiguity (0.15 $\rightarrow$ 0.28 %). Going tighter is expensive: ± 15 ns already drops 3 points.

What the missing 4 % is, and per arm



Left: the per-arm offset is reproducible between the two independent hours to ≤ 2.6 ns — an instrumental constant of the trigger path, so it belongs in the calibration. It is *not* a detector misalignment: step 1 sees the four wall flash times within 4 ns. **Right:** each arm takes ~ 24 % of the triggers, and no arm is anomalous.

DREAM triggers	213 420
matched, wall only	98.59 %
matched, wall AND plastic	96.00 %
plastic partner wall matched	97.38 %
plastic leg costs	2.58 %
wall leg costs	1.41 %

(Plateau values — a window wide enough that the window itself costs nothing, with the measured accidental rate subtracted.)

The plastic leg is the same ~ 2.6 % the v12 acceptance test reports, and it is the one number the reprocessing did not close — three very different plastic reconstructions give the same figure, so it is not a pulse-recognition problem. If it is worth chasing it is the discriminator threshold model, the channel selection, plastic dead time, or genuine detector inefficiency.

The calibration constants

Offline, DREAM→n_TOF time map

K	rate	1.103724×10^{-4}
T_0	offset	-253.64 ns
a_A	arm A	-16.81 ns
a_B	arm B	$+7.55 \text{ ns}$
a_C	arm C	$+1.62 \text{ ns}$
a_D	arm D	-0.83 ns
$\delta a_b, \delta k_b$	per bunch	fitted, ≥ 20 trig.
window	accept	$ r < 25 \text{ ns}$, one band

Offline, n_TOF internal

LIQ, PSS, WAL relative offsets	none needed
wall top—bottom pairing	measure per file
tflash repair on v12	off

DAQ, as operated in run 224572 (*not* fitted — read back from `n1081b_config.json` and required to emulate the trigger)

arm	wall thr.	plastic thr.
A	25 mV	118 mV
B	35 mV	139 mV
C	34 mV	157 mV
D	36 mV	134 mV

What transfers and what does not. The DAQ thresholds are hardware state: valid until someone turns a knob. The n_TOF-internal alignment is a property of the *processing* and must be re-measured per reprocessing. K , T_0 and the per-arm offsets are properties of the pair (DREAM run, n_TOF processing) — **re-fit them per run**; the per-bunch term is by construction per bunch and is always re-fitted.

Exported to
`nTof_x17_DAQ/calibrations/dream_ntof/`.

The method

1. Align `n_TOF` to itself against the beam pickup; confirm on late coincidences. No offsets turn out to be needed *on v12* — but check, do not assume.
2. Rebuild the N1081B sector SINGLES from the hit trees, with the thresholds the run actually held.
3. Fit the global time map (K , T_0 , per arm) on the whole run.
4. Fit $(\delta a_b, \delta k_b)$ per bunch on its own matched triggers.
5. Accept $|r| < 25 \text{ ns}$, single band.

Reproducible end to end from
`ntof_dream_merge/match_study/`; constants and the
per-run recipe in
`ntof_dream_merge/DREAM_NTOF_CALIBRATION.md`.

Why it matters for the Micromegas

- ▶ The MM cross-check keys on *which arm* fired. At $\pm 25 \text{ ns}$ the two-arm ambiguity is **0.15 %** and the accidental contamination of the matched sample is **0.049 %** — both directly limit how sharply an arm-vs-chamber correlation can be measured.
- ▶ A **6 ns** match residual means the `n_TOF` wall time can be used as an *absolute* time reference for the MM drift window. At the 30–40 ns level that was not possible.
- ▶ The liquid leg is already on the wall time base to under a nanosecond, so it can join without further calibration.